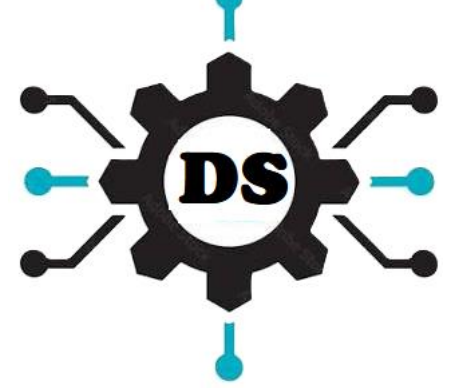




Introduction to Data Science **DTS 201**

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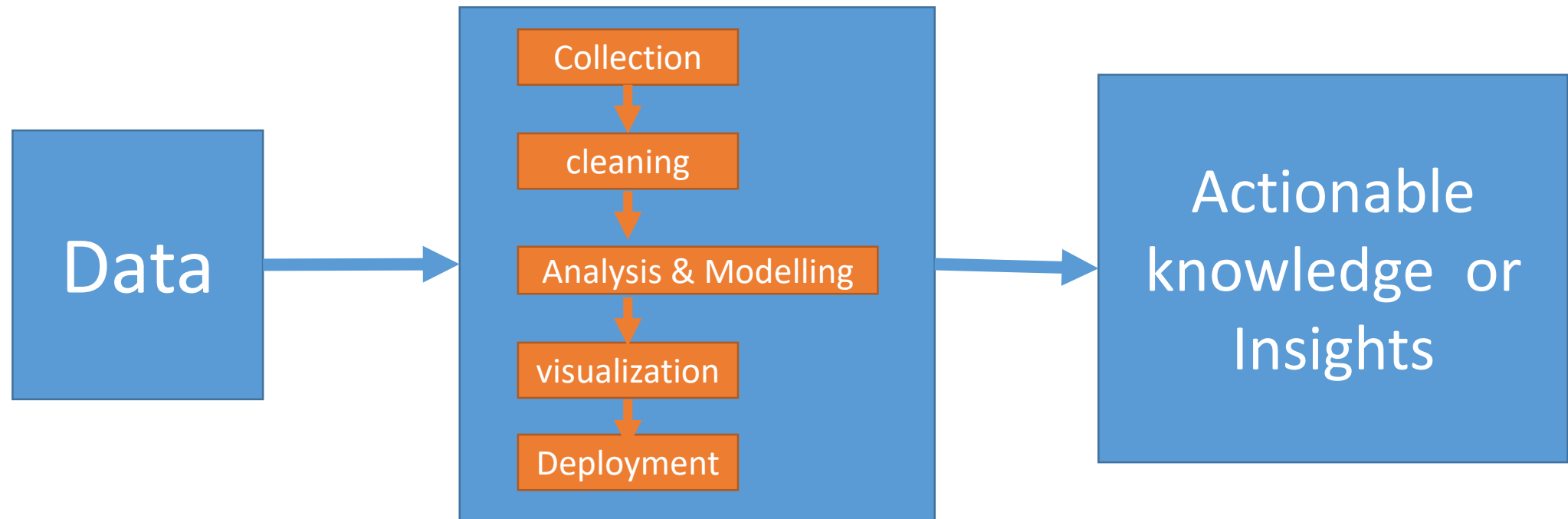
Week 1

Data Science Fundamentals

- What is Data Science and why is it important?
- Real-world applications and career pathways
 - Fundamentals of Data Science

What is Data Science and why is it important?

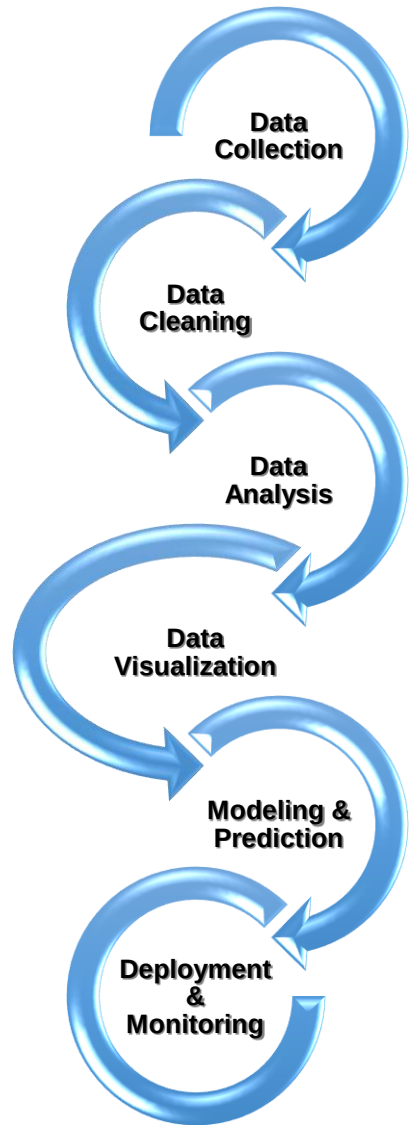
Data Science is the interdisciplinary field that uses mathematics, statistics, computer science, and domain expertise to extract meaningful insights from raw data, enabling predictions, decision-making, and innovation across industries.



Ultimate goal of Data Science: To **turn raw data into actionable intelligence**.

Data Science is the backbone of modern decision-making and innovation

Data Science Life Cycle



- ❑ **Data Collection** – Gathering raw data from sources like databases, sensors, or user interactions.
- ❑ **Data Cleaning** – Removing errors, duplicates, and inconsistencies to ensure accuracy.
- ❑ **Data Analysis** – Applying statistical and computational methods to identify relationships and trends.
- ❑ **Data Visualization** – Presenting findings through charts, dashboards, or interactive tools.
- ❑ **Modeling & Prediction** – Using machine learning algorithms to forecast future outcomes.
- ❑ **Deployment & Monitoring** – Implementing models in real-world systems and tracking performance

Importance of Data Science

Data Science is important because it transforms massive amounts of raw data into actionable insights, enabling smarter decisions, innovation, and efficiency across industries

1. **Enhances Decision-Making: Data Science provides data-driven decisions which outperform intuition-based ones. For example,** organizations use Data Science models such as predictive models to forecast trends, customer behavior, and risks. A Retailers can optimize inventory by predicting demand increase.
2. **Drives Innovation: DS creates new products and services that emerge from data insights.** Tech companies like Netflix and Amazon rely on recommendation systems powered by data science. Healthcare uses data science for drug discovery and personalized treatment.
3. **Improves Efficiency: DS provides Automation and optimization which reduces costs and save time. Examples:** Logistics firms use data science to optimize delivery routes. Today, Banks detect fraudulent transactions in real-time using DS skills
4. **Unlocks Hidden Patterns: DS can extract insights from Big data containing valuable but hidden signals. Example:** Data science provide early disease indicators such as the case of COVID-19 disease
5. **Impacts Daily Life: Data science shapes everyday experiences. E.g** Personalized ads, smart assistants, and even traffic navigation apps rely on Data science. Governments use Data Science for public health monitoring and resource allocation

Real-world applications

Industry	Data Science Impact
Healthcare	Disease prediction, drug discovery, patient monitoring
Finance	Fraud detection, credit scoring, algorithmic trading
Retail	Customer segmentation, recommendation engines
Transportation	Route optimization, autonomous vehicles
Government	Policy planning, public safety, resource allocation

Data Science career pathways

Data Science offers diverse and lucrative career paths, ranging from entry-level roles like Data Analyst to advanced positions such as Machine Learning Engineer and Chief Data Officer.

Role/Career Paths	Focus Area	Key Skills	Career Level
Data Analyst	Reporting & insights	SQL, Excel, Tableau	Entry-level
Data Scientist	Predictive modeling	Python, ML, Statistics	Mid-level
Machine Learning Engineer	AI model deployment	TensorFlow, PyTorch, Cloud	Mid/Senior
Data Engineer	Infrastructure & pipelines	Spark, Hadoop, AWS/GCP	Mid-level
BI Developer	Dashboards & reporting	Power BI, SQL, ETL	Entry/Mid-level
AI Researcher	Advanced algorithms	Deep learning, NLP, CV	Senior/Research
Data Architect	System design	Database, Cloud, Big Data	Senior
Chief Data Officer	Strategy & leadership	Governance, Business Strategy	Executive

Fundamentals of Data Science

1. **Domain Expertise:** Understanding the **context of the data** (e.g., healthcare, finance, retail) to Help in asking the right questions and interpreting results meaningfully. Example: A data scientist in healthcare must understand medical terminology and patient data structures.
2. **Mathematics & Statistics:** Provides the foundation for building predictive models and validating results using **knowledge in Probability & Statistics** (for hypothesis testing, regression, correlation, distributions) and **Linear Algebra & Calculus** (essential for machine learning algorithms).
3. **Programming Skills:** Enables automation, data manipulation, and model building using languages(Python, R, SQL), libraries (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), and applications (SPSS, MS Excel, Power BI, Orange, Weka, MATLAB)
4. **Data Management including Data Collection**(gathering structured and unstructured data), **Data Cleaning**(Handling missing values, duplicates, and inconsistencies), **Data Storage**(using databases (SQL, NoSQL) and big data frameworks such as Hadoop, Spark).
5. **Data Visualization:** representing data and converting complex results into **clear, visual insights** for stakeholders
6. **Machine Learning & AI Basics:** modelling and analyzing of data using Supervised Learning(Regression, classification), **Unsupervised Learning**(Clustering, dimensionality reduction) and **Deep Learning**(Neural networks for advanced tasks like image recognition and NLP).
7. **Data Ethics & Privacy:** ensuring fairness, transparency, and responsible use of data by compliance with regulations like GDPR and Nigeria Data Protection Act and avoiding bias in algorithms and protecting user privacy

Week 2

Understanding Data

- ☐ What is Data?
- ☐ Basic classification
- ☐ Data Examples from different domains
- ☐ Method of data collection in Data Science

What is Data?

Data is a form of facts, numbers, measurements, or symbols that can be collected, stored, and processed to provide meaning or insights.

Data are *collections of discrete or continuous values that convey information, describing quantities, qualities, facts, statistics or objects.*

Characteristics of Data

- ❑ **Raw form:** Data by itself may not have meaning until processed.
- ❑ **Types:** Can be qualitative (descriptive) or quantitative (numerical).
- ❑ **Digital nature:** In modern contexts, data is frequently stored and transmitted electronically.

Basic Data classification

Data are classified by Format, Nature, Sensitivity & Security, and Source

A. Data classification by format represents how the data are organized and stored.

- ❑ **Structured Data:** Highly organized and easily searchable in fixed fields. It is typically stored in relational databases (SQL). *Examples:* Spreadsheets, dates, names, and credit card numbers.
- ❑ **Semi-Structured Data:** Does not reside in a rational database but has some organizational properties like tags or markers that make it easier to process than raw data. *Examples:* XML, JSON files, and E-mails.
- ❑ **Unstructured Data:** Data that has no pre-defined format or organization. This makes up the vast majority (roughly 80%) of data generated today. *Examples:* Photos, videos, audio files, and social media posts.

Basic Data classification

B. Data classification by nature includes qualitative and quantitative data

- ❑ **Quantitative(Numerical):** Numerical data that can be measured or counted. Subtypes of Quantitative data:
 - **Discrete** (countable, e.g., number of students).
 - **Continuous** (measurable, e.g., height, weight).
- ❑ **Qualitative(Categorical):** Descriptive data that characterizes but does not measure. E.g Color (Blue), Texture (Smooth). Subtypes:

C. Data classification by Sensitivity & Security

- ❑ **Public Data** – Freely available, no restrictions (e.g., census reports).
- ❑ **Internal Data** – Limited to organizational use (e.g., employee directories).
- ❑ **Confidential Data** – Sensitive, requires protection (e.g., financial records, medical data).
- ❑ **Restricted Data** – Highly sensitive, strict access control (e.g., military intelligence).

D. Data classification by Source:

- ❑ **Primary Data** – Collected directly from original sources (surveys/questionnaires, experiments, interviews, field reports).
- ❑ **Secondary Data** – Derived from existing sources (reports, textbooks, online databases like Kaggle, UCI).

Data Examples from different domains

- ❑ **1. Healthcare Data:** Electronic Health Records (EHRs) (Patient demographics, diagnoses, treatments), Medical Imaging (X-rays, MRIs, CT scans), Genomic Data (DNA sequences, biomarkers), Wearable Devices(Heart rate, sleep patterns, activity levels).
- ❑ **2. Finance Data:** Transactional Data(Purchases, payments, transfers), Market Data(Stock prices, exchange rates, indices), Risk & Credit Data(Credit scores, loan histories), Fraud Detection Data(Suspicious transaction patterns)
- ❑ **3. Retail & E-commerce:** Customer Data(Purchase history, browsing behavior), Inventory Data(Stock levels, supply chain records), Sales Data(Revenue, discounts, seasonal trends), Recommendation Systems(Clickstream and preference data).
- ❑ **4. Education:** Student Records(Grades, attendance, demographics), Learning Analytics(Online course engagement, quiz scores), Institutional Data(Enrollment rates, faculty performance), Research Data(Surveys, experimental results).
- ❑ **5. Government & Public Policy:** Census Data(Population demographics, housing statistics), Economic Data(GDP, unemployment rates, inflation), Public Safety Data(Crime statistics, emergency response times), Environmental Data(Pollution levels, climate monitoring)
- ❑ **6. Technology & Social Media:** User Interaction Data(Likes, shares, comments),Network Data(Connections, communication patterns), IoT Data(Sensor readings from smart devices), Log Data(System performance, error tracking)
- ❑ **7. Transportation & Logistics:** GPS Data (Vehicle locations, routes), Traffic Data(Congestion levels, accident reports), Supply Chain Data(Shipping times, warehouse records), Autonomous Vehicle Data(Sensor inputs, driving decisions)

Method of data collection in Data Science

Data collection is the foundational step of the Data Science process

1. Primary Data Collection (Direct): Primary data is "first-hand" information collected specifically for the current research or project.

- ❑ **Web Scraping:** Using automated tools (like BeautifulSoup or Scrapy in Python) to extract data directly from websites. This is common for price monitoring or sentiment analysis.
- ❑ **APIs (Application Programming Interfaces):** Requesting data directly from a service provider (e.g., pulling tweets via the X/Twitter API or weather data from OpenWeatherMap). This is the most "stable" way to collect web data.
- ❑ **Surveys and Questionnaires:** Collecting qualitative and quantitative data directly from users. Tools like Typeform or Google Forms are often used.
- ❑ **IoT and Sensors:** Gathering real-time data from physical devices, such as temperature sensors, GPS trackers, or smartwatches.

Method of data collection in Data Science(continue)

2. Secondary Data Collection (Existing): Secondary data involves using information that has already been collected and organized by others.

- ❑ **Databases (SQL/NoSQL):** Querying internal company databases (like PostgreSQL or MongoDB) to retrieve historical records, customer transactions, or inventory logs.
- ❑ **Data Repositories:** Using public datasets from platforms like **Kaggle**, **UCI Machine Learning Repository**, or government portals (e.g., data.gov).
- ❑ **Data Warehouses & Lakes:** Accessing massive, integrated datasets from cloud platforms like Snowflake, AWS Redshift, or Google BigQuery.

Method of data collection in Data Science(continue)

3. Simulated data : **Simulated Data** is a specific method used when real-world data is scarce, expensive, or restricted by privacy laws. Simulated Data are generated programmatically rather than collected from real-world events. It mimics the statistical properties of real data. Simulated data are useful for:

- ❑ **Privacy:** To train models on sensitive topics (like medical records) without using real patient identities.
- ❑ **Edge Cases:** To "show" a model rare events that don't happen often in real life, like a specific type of engine failure.
- ❑ **Testing:** To create a controlled environment to see how a model reacts to specific variables.

Common tools used for simulating data are Monte Carlo simulations, Synthetic Data Vault (SDV), and Generative Adversarial Networks (GANs).

Week 3

Scope & Methodology

- Scope of Data Science in modern industries,
- Methodology of extracting knowledge from big datasets
- Overview of tools and platforms (Python, R, SQL, cloud platforms)

Scope of Data Science in modern industries

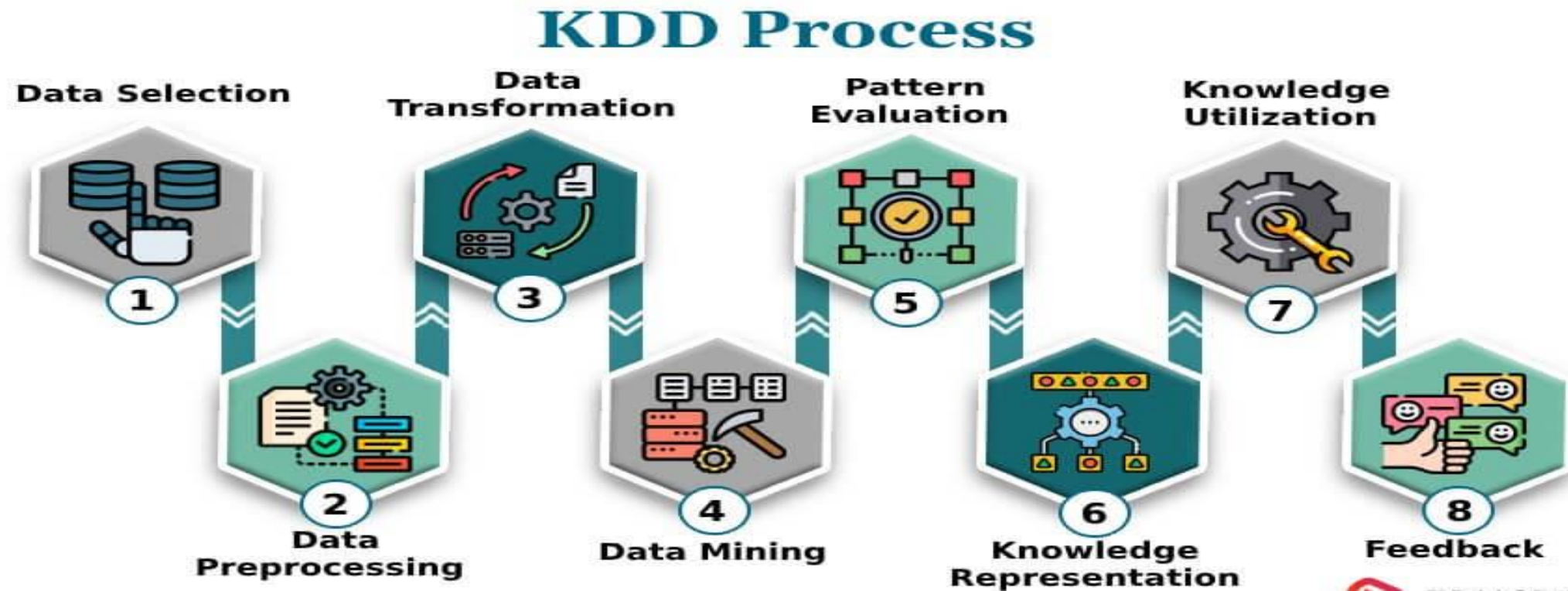
Data Science is a critical driver of innovation and efficiency across modern industries, powering everything from personalized healthcare to fraud detection in finance. Its scope is expanding rapidly, with applications in nearly every sector, making it one of the most in-demand fields globally. Data science provides **Universal applicability knowledge**(Every industry generates data, and data science helps transform it into actionable insights), creates **Competitive advantage** (Companies that leverage data science outperform those relying only on intuition), and facilitate **Future industrial growth** (world Economics are driven by technologies powered by data science that projects millions of new jobs in data science and AI in the nearest future).

Scope of Data Science in modern industries

Industry	Applications of Data Science	Impact
Healthcare	Disease prediction, drug discovery, patient monitoring	Improves patient outcomes, reduces costs
Finance	Fraud detection, credit scoring, algorithmic trading	Enhances security, optimizes investments
Retail & E-commerce	Recommendation systems, customer segmentation, demand forecasting	Boosts sales, personalizes shopping
Manufacturing	Predictive maintenance, supply chain optimization	Reduces downtime, increases efficiency
Transportation & Logistics	Route optimization, autonomous vehicles, fleet management	Cuts costs, improves safety
Education	Learning analytics, adaptive learning platforms	Enhances student performance
Government & Public Policy	Census analysis, resource allocation, crime prediction	Improves governance and planning
Technology & Social Media	Sentiment analysis, personalization, AI-driven features	Drives engagement and innovation

Methodology of extracting knowledge from big datasets,

Extracting knowledge from big datasets involves a structured methodology known as Knowledge Discovery in Databases (KDD). It includes steps such as data selection, cleaning, transformation, mining, and interpretation, often supported by machine learning and statistical techniques to uncover meaningful patterns and insights.



1. **Data Selection:** Focuses on data that aligns with the problem or business objective. Identify relevant datasets from multiple sources (databases, sensors, social media, etc.).
2. **Data Preprocessing (Cleaning & Integration):** Handles missing values, duplicates, and inconsistencies. Useful for data integration from different sources into a unified format. Example: Combining hospital records with wearable device data in healthcare.
3. **Data Transformation:** Converting raw data into suitable formats for analysis. It uses techniques such as normalization, aggregation, dimensionality reduction (e.g., PCA). It helps reduce complexity and improve efficiency.
4. **Data Mining:** Apply algorithms to discover patterns, correlations, and trends. Methods include: **Classification** (predicting categories, e.g., fraud detection), **Regression** (predicting continuous values, e.g., stock prices), **Clustering** (grouping similar data points, e.g., customer segmentation) and **Association Rule Mining** (finding relationships, e.g., “customers who buy X also buy Y”).
5. **Pattern Evaluation:** Assess discovered patterns for relevance, accuracy, and usefulness. Useful for Removing redundant or irrelevant results. Example: Filtering out spurious correlations in financial data.
6. **Knowledge Representation:** Present insights in understandable formats (dashboards, charts, reports). It requires the use of Visualization tools like Tableau, Power BI, or Matplotlib make results actionable.
7. **Knowledge Utilization:** Applying acquired knowledge—whether from data, research, or experience—into practical actions, decisions, and innovations
8. **Feedbacks:** Getting responses from stakeholders/users of acquired knowledge to improve a system or process

Overview of tools and platforms (Python, R, SQL, cloud platforms)

Data Science tools and platforms provide the backbone for handling big datasets, building models, and deploying insights.

A. Tools in Data Science: These are the everyday instruments data scientists use for analysis, modeling, and visualization:

- ❑ **Programming Languages: Python:** Most widely used, with libraries like Pandas, NumPy, Scikit-learn, TensorFlow.), **R**(Strong in statistical analysis and visualization and **SQL**(Essential for querying and managing structured data).
- ❑ **Data Visualization Tools: Tableau, Power BI** (provide business-focused dashboards), **Matplotlib, Seaborn**(Python libraries for charts and plots)
- ❑ **Machine Learning & AI Frameworks: TensorFlow, PyTorch**(for Deep learning frameworks), **Scikit-learn**(Classical machine learning algorithms)
- ❑ **Big Data Tools: Apache Hadoop, Spark**(Distributed computing for massive datasets), **Google BigQuery** (Cloud-based big data analytics)

B. Platforms in Data Science: Platforms integrate multiple tools into one ecosystem, making collaboration and deployment easier

Platform	Key Features	Best For
Anaconda	Package management, Jupyter notebooks, Python/R support	Individual data scientists, education
Databricks	Unified analytics, Spark integration, ML lifecycle management	Enterprise big data & AI projects
Google Cloud AI	Pre-trained ML APIs, scalable infrastructure	Cloud-native AI development
AWS SageMaker	End-to-end ML model building, training, deployment	Enterprise ML workflows
Microsoft Azure ML	Drag-and-drop ML, integration with Power BI	Business-focused ML solutions
KNIME & RapidMiner	Visual workflows, easy integration	Beginners and non-programmers

Week 4

Data Science Process I

- ❑ Data collection methods (surveys, sensors, APIs, databases)
- ❑ Data preprocessing: cleaning, normalization, handling missing values
 - ❑ Lab: prepare a sample dataset

Week 5

Data Science Process II

- Training and testing datasets
- Splitting data for model development
- Case study: simple supervised learning workflow

Week 6

Rudiments of Data Visualization

- Principles of effective visualization, Tools: Matplotlib, Seaborn, Plotly
- Lab: visualize distributions and relationships in a dataset

Course Outline

Week 7 – Probability, Distributions & Simulations: Probability fundamentals, Common distributions (normal, binomial, uniform), Simulations for data science applications

Week 8 – Predictions & Models: Introduction to predictive modeling, Regression and classification basics, Model evaluation metrics (accuracy, precision, recall, F1 score)

Week 9 – Domain Use Cases: Image data (computer vision basics), Natural language (text mining, sentiment analysis), Audio and video applications

Week 10 – Knowledge Extraction: Data mining fundamentals, Business Intelligence (BI) concepts, Knowledge management systems

Week 11 – Big Data & Analytics: Introduction to Big Data integration and intelligence (Hadoop, Spark), Introduction to Data Analytics (descriptive, diagnostic, predictive, prescriptive), Lab: analyze a medium-sized dataset

Week 12 – Programming & Capstone: Introduction to programming for data science (Python basics, libraries), Nini project: apply the full data science pipeline to a dataset, Present findings with report and visualization